Colorado State University

Series Hybrid Design

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Overview

- Series Hybrid Architecture
- Component Sizing
- Takeaway/Conclusions

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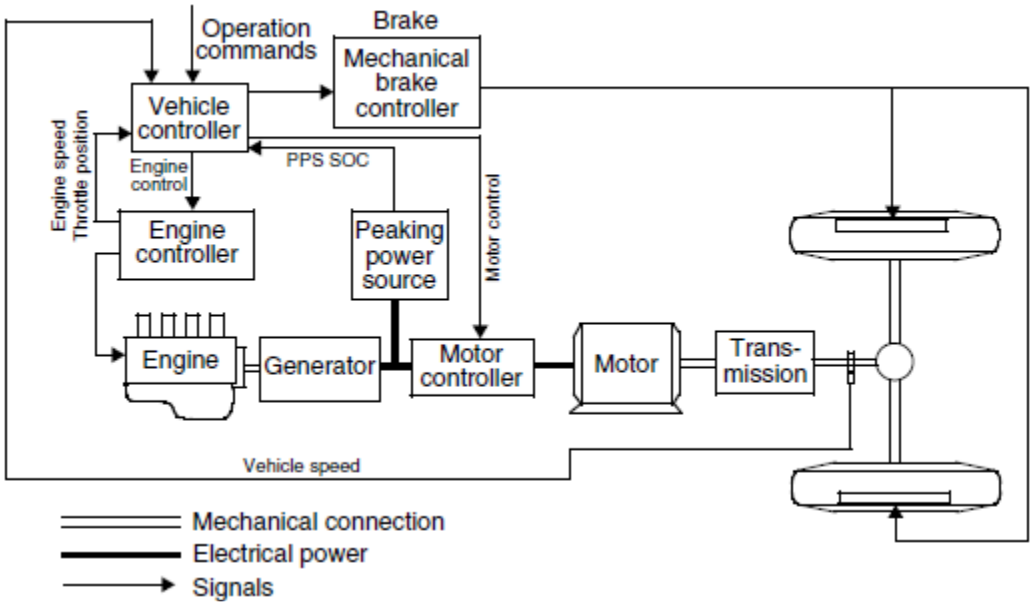
Series Hybrid

- Recall that a series hybrid is an all-electric powertrain, where the battery is recharged by an engine/generator



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Component Sizing

- To specify the series hybrid we have to specify the:
  - Of course, battery/motor/generator must be electrically (voltage and current) compatible
  - Of course, engine and generator must be mechanically (speed, torque) compatible
  - Motor torque and power
  - Generator torque and power
  - Battery power and energy capacity
  - Engine torque and power

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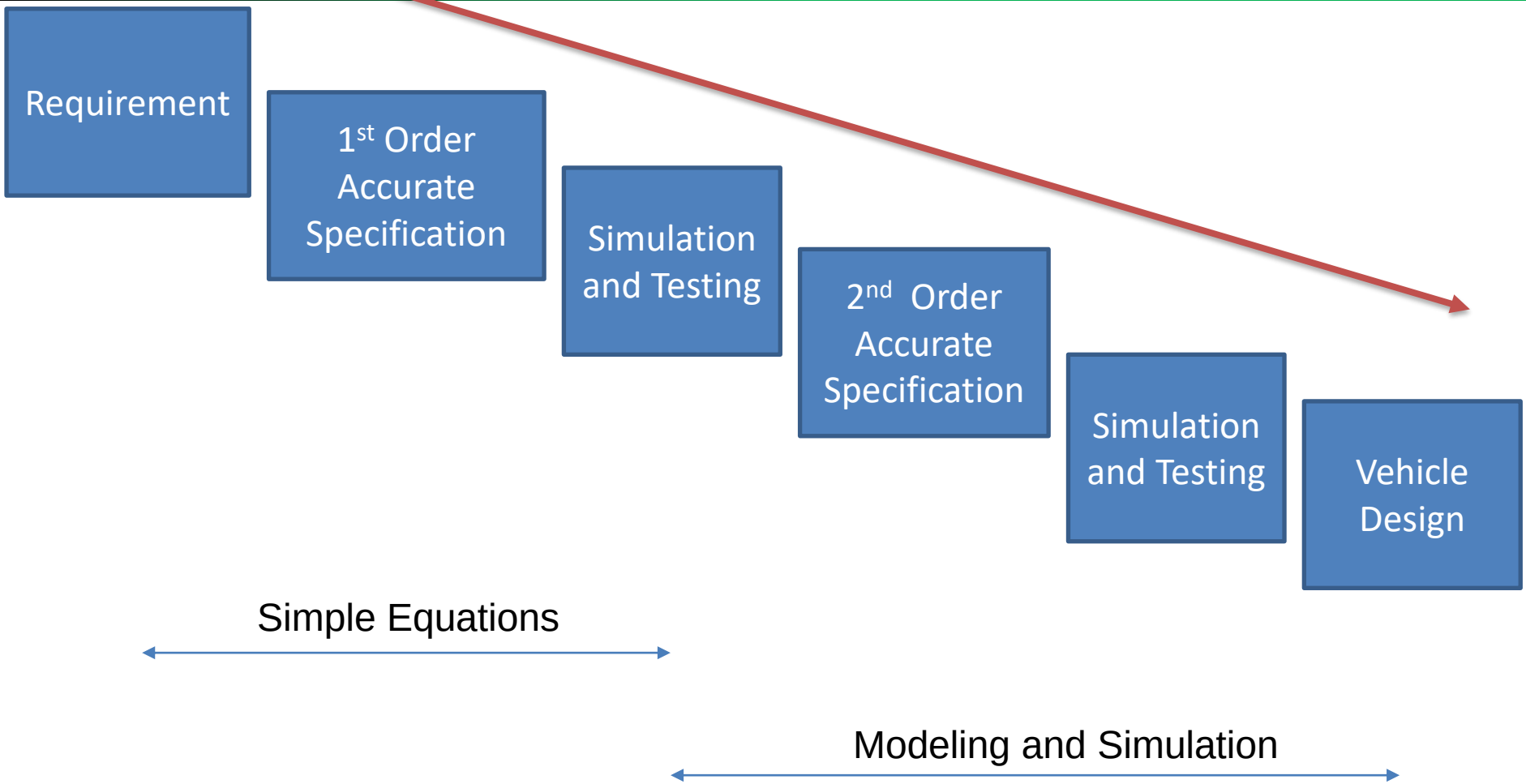
Component Sizing for this Course



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Requirement Set for Series Hybrid

- Medium sized vehicle
  - 0-60mph in 8 sec
  - Top speed 100 mph
  - No ZEV range
  - Gradability of 6% at 60mph
  - Tires = 205/65-15

20.5 cm / 2.54 = 8.070 in

8.070 in x 0.65 = 5.245 in

15 in + 2 * 5.245 in = 25.49 inches in Diameter

r_{wh} = 0.323 meters

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1st Order Motor Sizing

- Given an acceleration requirement, the performance of the series HEV is determined by the electric motor power

$$P_t = \frac{\delta M_v}{2t_a}(V_f^2 + V_b^2) + \frac{2}{3}M_v g f_r V_f + \frac{1}{5}\rho_a C_D A_f V_f^3,$$

```
clear
close all

mv = 1623 ;%kg,  mass of vehicle
delta = 1.04 ;% equivalent mass factor of rotating inertia
ta = 8 ; %seconds, time required for acceleration
Vf = 27 ;% m/s, final velocity
Vb = 0; % m/s, beginning velocity
g = 9.81  ; % m/s/s, accel due to gravity
fr = 0.010; % unitless, Crr, coeff of rolling resistance
rho = 1.2; % kg/m3, density of air
Cd = 0.29; % unitless, Cd, coeff of drag
Af =2.3 ; % m2, frontal area of the vehicle

Pt = delta*mv/2/ta*(Vf^2 + Vb^2) + ...
    2/3*mv*g*fr*Vf + 1/5*rho*Cd*Vf^3
```

Pt >= 82 kW

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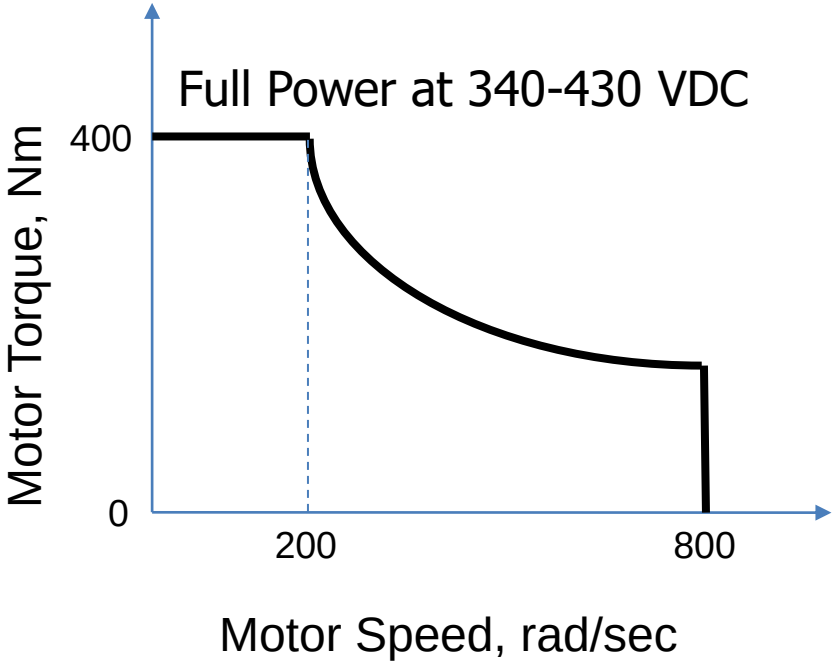
1st Order Motor Sizing

- Consider a 1-spd transmission to meet the top speed requirement

$$V_{max} = r_{wh} * i_{tx} * \omega_{mot}$$

$44.7 m/s = 0.36 * i_{tx} * 800$

$i_{tx} = 5.7$



https://www.neweagle.net/support/wiki/docs/Datasheets/UQM/PP145.pdf

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1st Order Motor Sizing

- My 0 ZEV range means that the battery does not have to have any particular amount of energy
- I'd still like the vehicle to be able to accelerate without starting the engine, so the battery must be able to source >80kW.
- I'll choose a "high power" and therefore low energy cell
- http://www.a123systems.com/Collateral/Documents/English-US/A123%20Systems%20ANR26650%20Data%20Sheet.pdf

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1st Order Batt Sizing

- For chosen motor, Full Power at 340-430 VDC
- Assume that maximum pulse discharge is close to matched impedance current ½*3.3V/cell = ~1.7 V/cell
- I need 340V*1.7V/cell = 200 cells in series
- 80kW @ 340V = 235A, so I need 2 parallel strings
- Batt cell mass 0.076kg*400 = 30kg
- Batt_energy = 2.4Ah/cell*3.3V/cell*400cells = 3.1kWh
- Usable battery energy might be 60% of that

ANR26650M1-B Cell Specifications	
Cell Dimensions (mm)	ø26 x 65
Cell Weight (g)	76
Cell Capacity (nominal/minimum, Ah)	2.5/2.4
Voltage (nominal, V)	3.3
Internal Impedance (1kHz AC typical, mΩ)	6
HPPC 10 Sec Discharge Pulse Power 50% SOC	200 W
Recommended Standard Charge Method	1C to 3.6V CCCV, 45 min
Recommended Fast Charge Method to 80% SOC	4C to 3.6V CC, 12 min
Maximum Continuous Discharge (A)	70
Maximum Pulse Discharge (10 seconds, A)	120
Cycle Life at 10C Discharge, 100% DOD	>1,000 cycles
Operating Temperature	-30°C to 55°C
Storage Temperature	-40°C to 60°C

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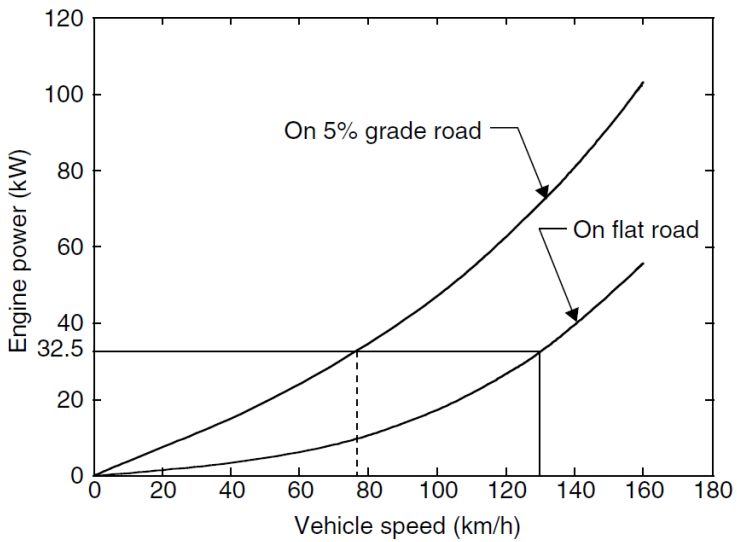
1st Order Generator Sizing

- Because my battery has very little energy, I will need to have the generator and engine provide the power to climb grades

```
clear
close all

mv = 1623 ;%kg,  mass of vehicle
delta = 1.04 ;% equivalent mass factor of rotating inertia
ta = 8 ; %seconds, time required for acceleration
Vf = 27 ;% m/s, final velocity
Vb = 0; % m/s, beginning velocity
g = 9.81  ; % m/s/s, accel due to gravity
fr = 0.010; % unitless, Crr, coeff of rolling resistance
rho = 1.2; % kg/m3, density of air
Cd = 0.29; % unitless, Cd, coeff of drag
Af =2.3 ; % m2, frontal area of the vehicle
Beta = 0.05; % 5% grade
Elec2WheelEff = 0.8; % efficiency from gen elect to wheels

Pgen = 1/Elec2WheelEff* Vf*(1/2*rho*Vf*Vf*Af*Cd + mv*g*fr +
mv*g*Beta )
```



Pgen >= 42kW

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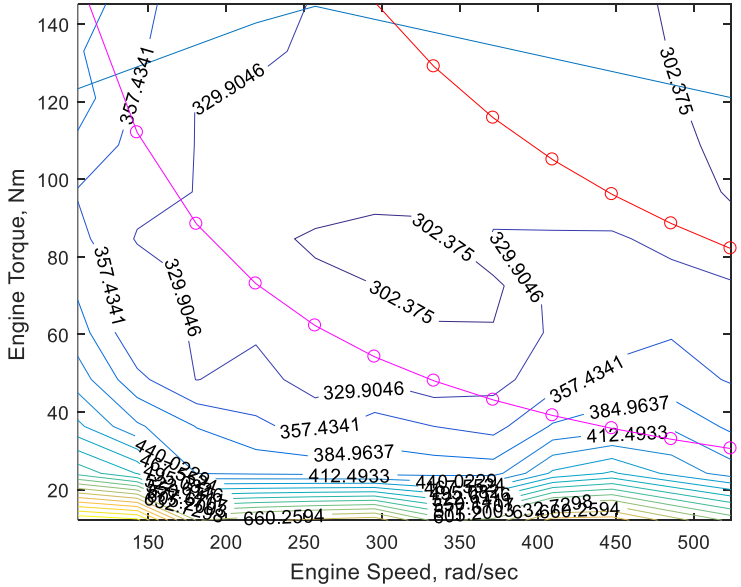
1st Order Engine Sizing

- The engine will need to generate at least 42kW, I choose 63kW

```
c_rpm_radps = (2*pi)/60;
eng_si_1900_63_saturn_HOT
figure; plot(eng_spd_max_hot_index, eng_trq_max_hot_map)
xlabel('Engine Speed, rad/sec'); ylabel('Engine Torque, Nm')
bsfc_map = (eng_fuel_hot_map*3600*1000) ./ ((eng_spd_fuel_hot_index' * eng_trq_fuel_hot_index)/1000)
hold on  ; [c,h]=contour(eng_spd_fuel_hot_index, eng_trq_fuel_hot_index,bsfc_map', 20); clabel(c,h)
a = axis;
Power_Line_Plotted = 43e3; %plot this power on the engine map
plot(eng_spd_max_hot_index, Power_Line_Plotted./eng_spd_max_hot_index,'ro-')
axis(a)
Power_Line_Plotted = 16e3; %plot this power on the engine map
plot(eng_spd_max_hot_index, Power_Line_Plotted./eng_spd_max_hot_index,'mo-')
axis(a)
```



- Good combination of engine maximum power, and engine efficiency during “normal” operation

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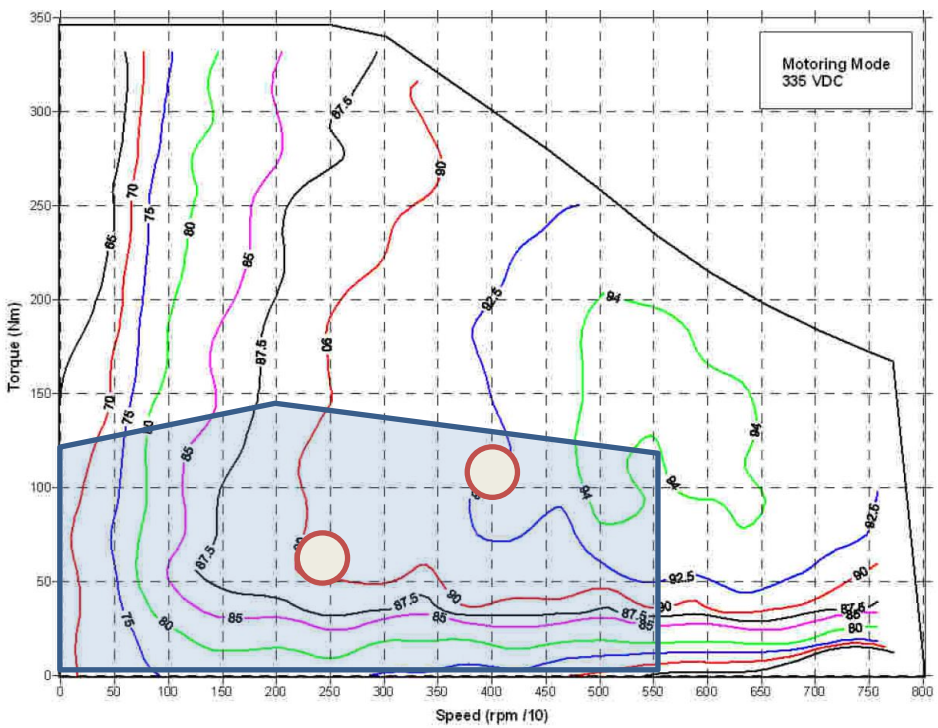
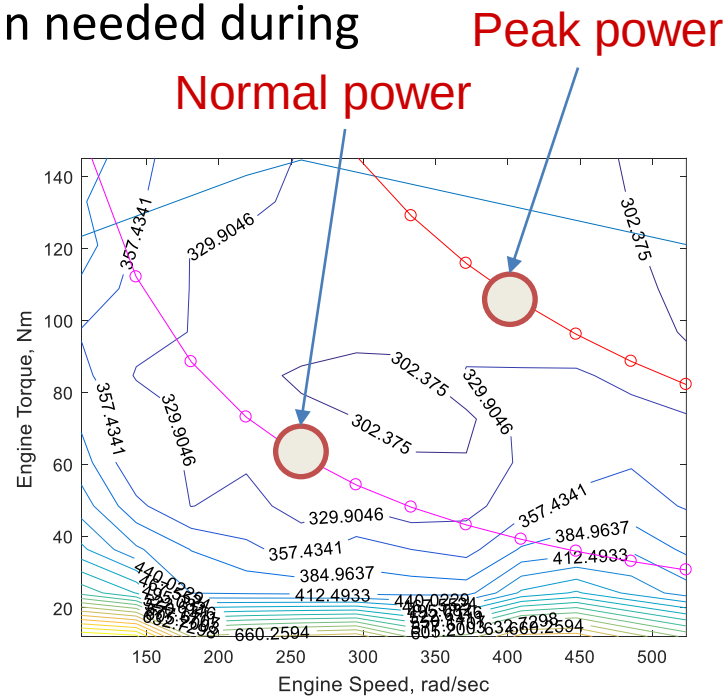
1st Order Generator Sizing

- I need a motor that is electrically and mechanically compatible with the vehicle as designed.
- https://www.neweagle.net/support/wiki/docs/Datasheets/UQM/PP135.pdf
- 60kW continuous power, which is more than needed during normal or peak power





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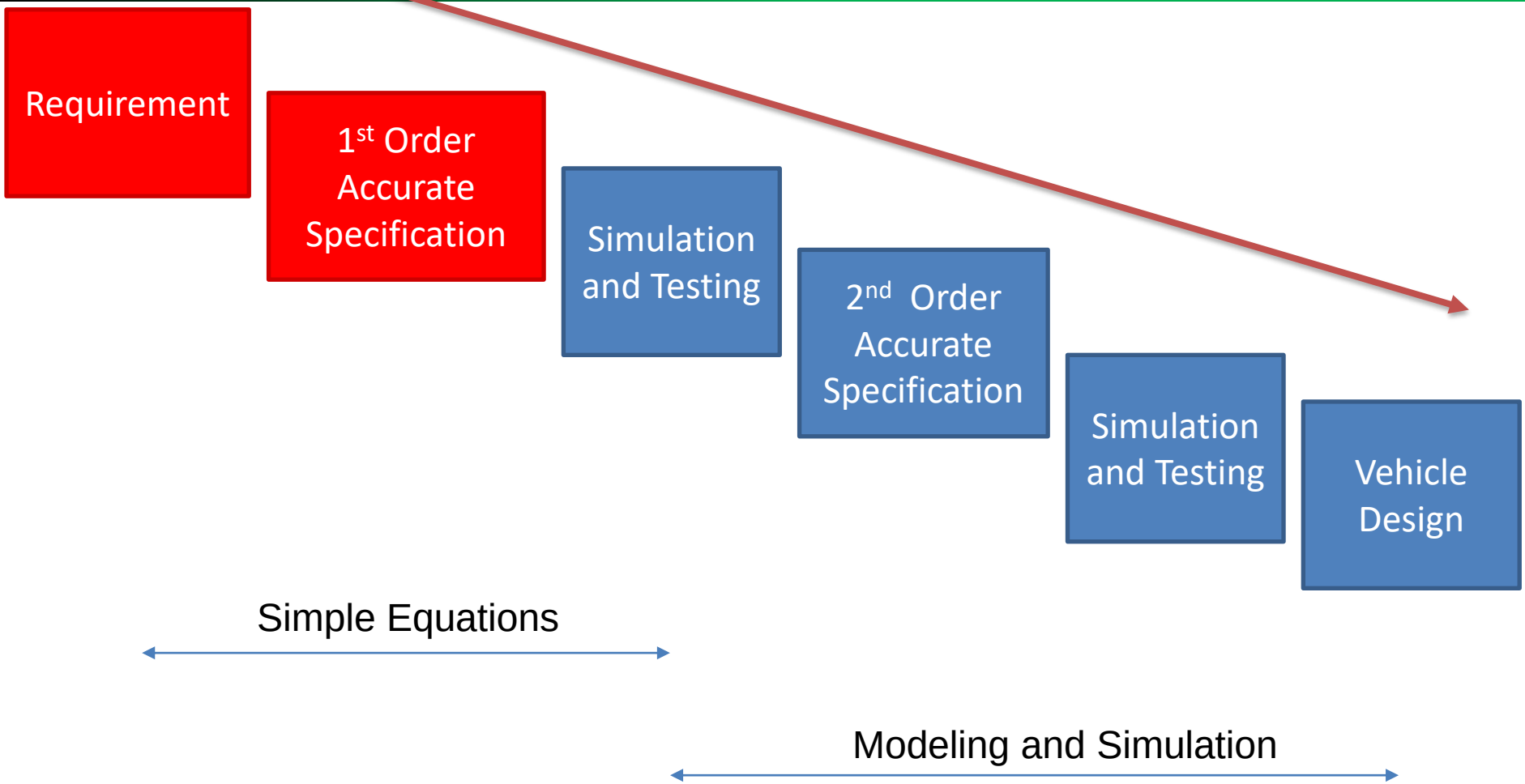
Component Sizing for this Course



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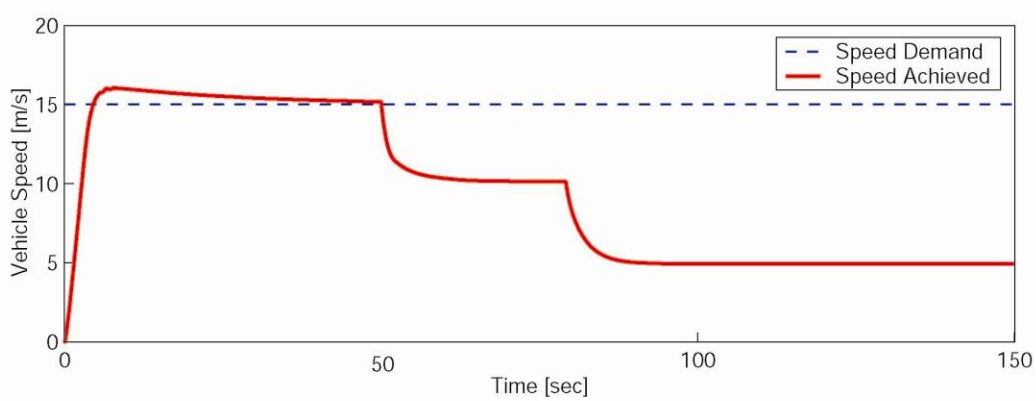
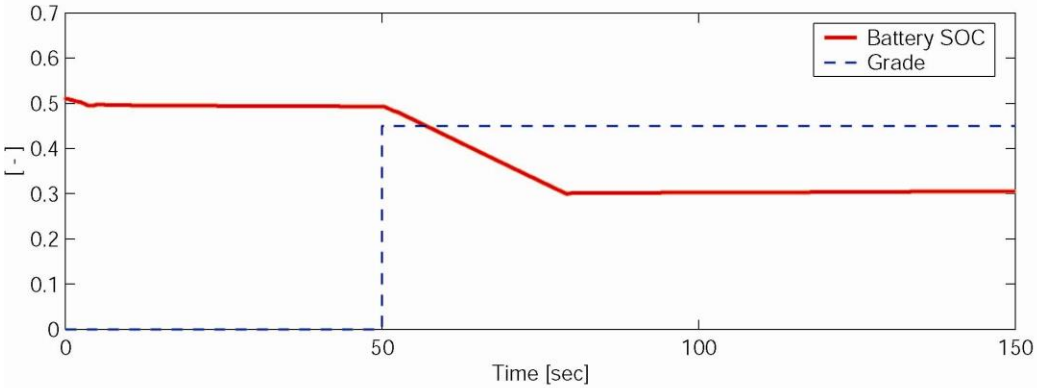
Simulation and Testing

- Assemble this in your Simulink model
- Develop baseline controls
- Evaluate your vehicle against the requirements





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Simulation and Testing

- Assemble this in your Simulink model
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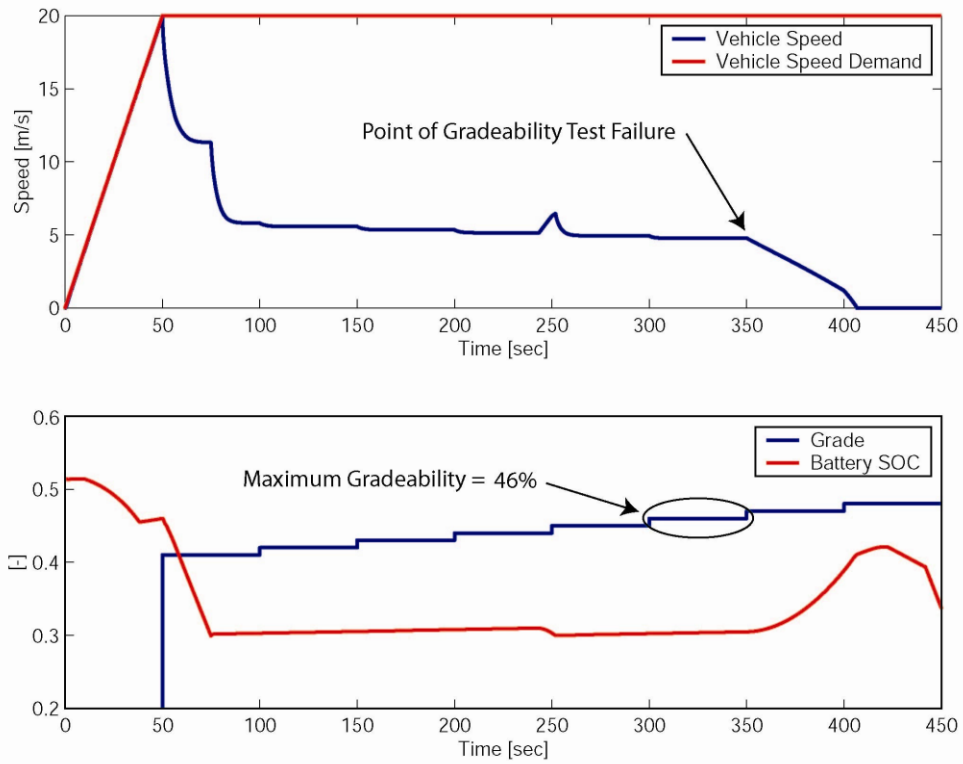

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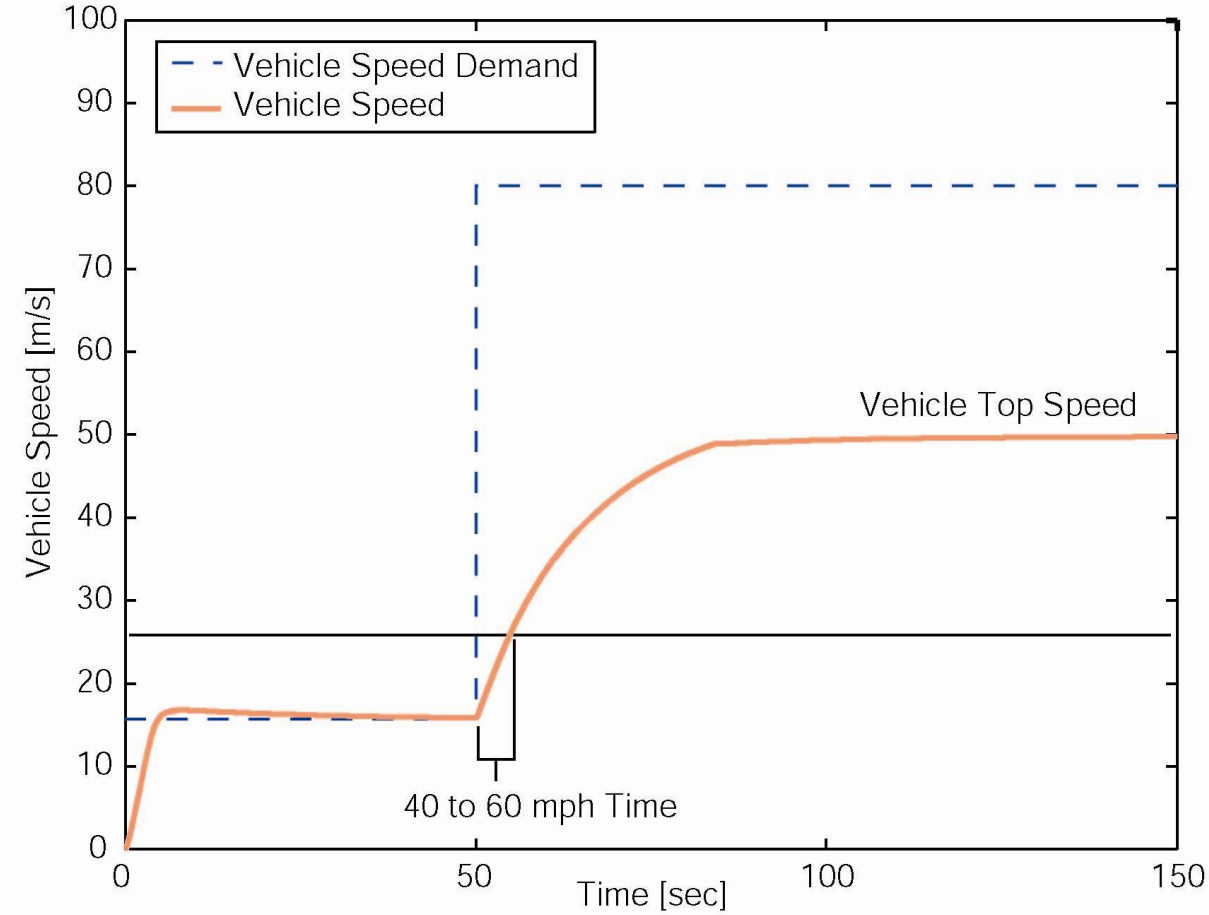
Simulation and Testing

- Acceleration Testing



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Takeaway/Conclusions

- What would be the requirements for
  - Top speed
  - Gradability @ 60mph
  - Gradability at 2 mph
  - Fuel economy
  - 0-60 time

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